# Root Cause Analysis Report

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|---|---|
| Part | MS-4505-001 Sheet Metal Bracket |
| Material / process | SPCC Steel / bending |
| Case | rca_9438444eb67d422ba694823dfc55aa4f |
| Date | 2026-11-05 15:57:56 |
| Status | AI proposal, pending expert review |

**WARNING:** the AI model was not used for this analysis (fake-offline). The text below is a generic offline template, not a result.

## 1. Problem

| Defect | Severity | Confidence | Detail |
|---|---|---|---|
| dent | major | 70% | |
| crack | critical | 59% | |
| crack | critical | 56% | |
| scratch | minor | 38% | |
| crack | critical | 31% | |
| crack | critical | 30% | |
| crack | critical | 24% | |
| dent | major | 22% | |
| scratch | minor | 20% | |
| scratch | minor | 20% | |
| scratch | minor | 16% | |
| scratch | minor | 10% | |

## 2. Root cause

**Cause:** Improper Bending Force / Over Bending

**Reason:** The detected deformation and dimensional inaccuracy strongly suggest excessive or improperly calibrated bending force. Retrieved RCA documents confirm this pattern for SPCC steel bending operations.

**Effect:** Deformation, Dimensional Inaccuracy

**Confidence:** 61%

**Supporting evidence**

- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba
- **587aa665-c6f1-48df-8783-09d65ff4f82a**: Tool Wear / Die Damage Root Cause Analysis Defect Types: scratch, dent, edge burr, dimensional shift Process: bending, stamping, forming Root Cause: Progressive wear of punch/die tooling causes dimensional drift, surface scratches, edge burrs and dents on formed parts. Effect: Surface defects (scratches from worn tooling edges), dents from damaged die surfaces, edge burrs from worn cutting edges,
- **6872e1ff-9d0f-41f8-9cd9-c783d66c1b46**: Springback Root Cause Analysis Defect Types: deformation, dimensional inaccuracy Process: sheet metal bending Root Cause: Elastic recovery of sheet metal after forming (springback) causes the bent angle to open up beyond the intended angle. Effect: Dimensional inaccuracy in bent angles, parts out of tolerance on angular dimensions and related linear dimensions. Corrective Actions: 1. Apply overben

## 3. Solution

### *Corrective actions (fix this occurrence)*

1. Re-set the press-brake bending force and angle to the material specification
2. Re-inspect the affected lot for deformation and dimensional deviation

### *Preventive actions (stop it recurring)*

1. Add bending-force verification to the machine calibration schedule
2. Control bending parameters per material batch

### *Precautions*

- Optimize bending parameters to material specification
- Use proper tooling and punch/die sets
- Regular machine calibration and force verification

### *How to verify the fix*

Re-inspect the next 30 parts; no deformation and all dimensions within tolerance

## 4. Other causes considered

| Rank | Cause | Confidence | Reason |
|---|---|---|---|
| 2 | Material Inconsistency / Hardness Variation | 58% | Hardness variation in the incoming material can cause unpredictable springback and cracking. Material certificate verification is required. |
| 3 | Tool Wear / Die Damage | 55% | Progressive die wear causes scratches, dents, and edge burrs. Preventive maintenance schedule should be implemented. |

## 5. Traceability

AI model: fake-offline. Expert edited the cause: no.
*This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.*